

MSc Computing 2026-27 Pre-sessional Activities

We look forward to welcoming you to the MSc Computing programme!

C++ Programming

You will be spending the first week studying C++ programming intensively as part of the *Principles and Practice of Programming* module. If you would like a preview of the first part of the module, please see <https://www.doc.ic.ac.uk/~wjk/C++Intro> [Note: the content is subject to change before the term starts]

We **strongly recommend** that you familiarise yourself with some of the basics before the module starts. Doing this will help you keep up with the fast pace of the module. The recommended textbook is:

Walter Savitch, [Problem Solving with C++: Global Edition](#) (10th edition), Pearson Education, January 2018. ISBN 978-1292222820.

[The online version](#) is available once you have received access to the library.

At a minimum, we suggest that you read the following sections:

<i>Chapter 1</i>	Computer Systems	1.1, 1.2, 1.3, 1.4
<i>Chapter 2</i>	C++ Basics	2.1, 2.2, 2.3, 2.4, 2.5
<i>Chapter 4</i>	Procedural Abstraction and Functions	4.1, 4.2, 4.3, 4.4, 4.5, 4.6

An integral part of learning programming is to write, test, and debug your own programs. It is not enough to just read! Previous cohorts have found that the more practice at programming they have before arriving, the better they cope with the module. Therefore, if you have access to a C++ compiler (e.g. g++ under Linux or Microsoft Visual C++ under Windows), you should **attempt** one or two of the Self-Tests and Programming Projects in Chapters 1, 2 and 4.

The following sections contain more advanced concepts that would be helpful, but not essential, to read:

<i>Chapter 3</i>	More flow of control	3.1, 3.2, 3.3, 3.4
<i>Chapter 5</i>	Functions	5.1, 5.2, 5.3
<i>Chapter 7</i>	Arrays	7.1, 7.2
<i>Chapter 9</i>	Pointers	9.1

Linux

You will be using Linux shell commands during your degree, especially in the *Principles and Practice of Programming* and *Software Systems Engineering* modules. You may also need to run experiments remotely on the lab machines.

It is thus worth being familiar with the basics of the Linux operating system. You can follow an example introductory course at <https://www.doc.ic.ac.uk/~wjk/UnixIntro>. If you are short on time, focus on Chapters 2, 3, 4 & 6.

The [lab machines in the department](#) will run on **Ubuntu 26.04 LTS** in the 2026/27 academic year, so you may find it easier to use the same distribution, although this is not necessary.

If you are using the latest version of Windows 11, you can install [a virtual Ubuntu distribution via Windows Subsystem for Linux \(WSL\)](#). Alternatively, you can install [a Ubuntu distribution on VirtualBox](#) for a full Ubuntu virtual desktop.

Operating System on your own machine

You are free to use whatever Operating System you prefer on your own machine. You can connect remotely to lab machines via SSH so it is not necessary for you to use a Linux distribution on your own machine. There will be a lab session in the first week on how to connect remotely to lab machines.

Mathematics

If you had limited or no exposure to mathematics—particularly calculus, linear algebra, or probability and statistics—during your undergraduate studies, we recommend taking the elective module *Mathematics for Computing* in the Autumn term. The module is designed to help you build a sufficient foundation for specialised modules such as *Machine Learning*, *Computer Vision*, *Graphics*, and *Algorithms* in the Spring term. Note that the module was previously known as *Logic and Mathematics for Computing*. The *Logic* part will be moved to what will be called the [Logic and Symbolic AI](#) module which will be offered as an elective module to you in 2026/27.

The module assumes that you have had a decent grounding in mathematics at secondary school level. If you have not studied mathematics for some time, we recommend revisiting topics typically covered at that level to help you engage effectively with the content and pace of the module.

We encourage you to:

- Refer to the syllabus of the **mathematics** part of the existing [module specification](#), and
- Refresh the initial key topics of the syllabus using standard A-level textbooks, such as Edexcel's [Pure Mathematics Year 1](#), [Pure Mathematics Year 2](#), [Statistics Year 1](#), and [Statistics Year 2](#). Equivalent materials for the International Baccalaureate (IB) in Mathematics, [Khan Academy](#), or other reputable online resources may also be beneficial.
- Go through the slide deck on Foundation Mathematics via this [Google Drive link](#). You may find it helpful to go through these well in advance before the start of the mathematics part of the module.

The main textbook for the module is: *Glyn James and Phil Dyke, Modern Engineering Mathematics* (6th Edition), Pearson Education, 2020. ISBN 9781292253497. [The online version](#) is also available once you have received access to the library.

See you at Imperial College London in the autumn!